Mini-project 6.3

Customer Segmentation Analysis Report

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To Marketing Stakeholders

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Introduction

The organization currently manages a massive global operation covering 47 countries, but our marketing approach is still too broad and fragmented. Even with nearly a million rows of transaction data at our fingertips, we aren't fully tapping into the unique stories those numbers tell about our customers' habits.

The major challenge is that our current data is just a collection of orders—it doesn't yet show us who our customers actually are. Because we haven't grouped these customers into clear segments, we risk wasting marketing resources on "one-size-fits-all" campaigns that miss the mark. For example, we might treat a loyal, frequent shopper the same way we treat a one-time customer, which leads to missed opportunities for growth and retention.

Furthermore, working with such a large, international dataset brings up real concerns regarding data quality and the need for ethical, unbiased analysis. We need to bridge this gap by turning our raw data into five distinct, actionable customer profiles—balancing technical accuracy with a practical strategy that our marketing teams can actually use to drive engagement.

Scope of Analysis

Information at our disposal show that the data is composed of 951,668 rows of transaction records with 20 features. Managing such enormous data required a meticulous approach to data quality and preparation. To build an accurate model, we focused our feature engineering on five strategically selected metrics that best capture customer value and behavior:

- Recency & Frequency: To measure engagement and brand loyalty.
- Customer Lifetime Value (CLV): To identify our most significant long-term contributors.
- Average Unit Cost: To distinguish between budget-conscious and premium shoppers.
- Customer Age: To provide a vital demographic layer to our behavioral findings.

Methodology

To transform this vast information into actionable intelligence, we employed a series of data science workflows as presented in figure 1 and table 1.

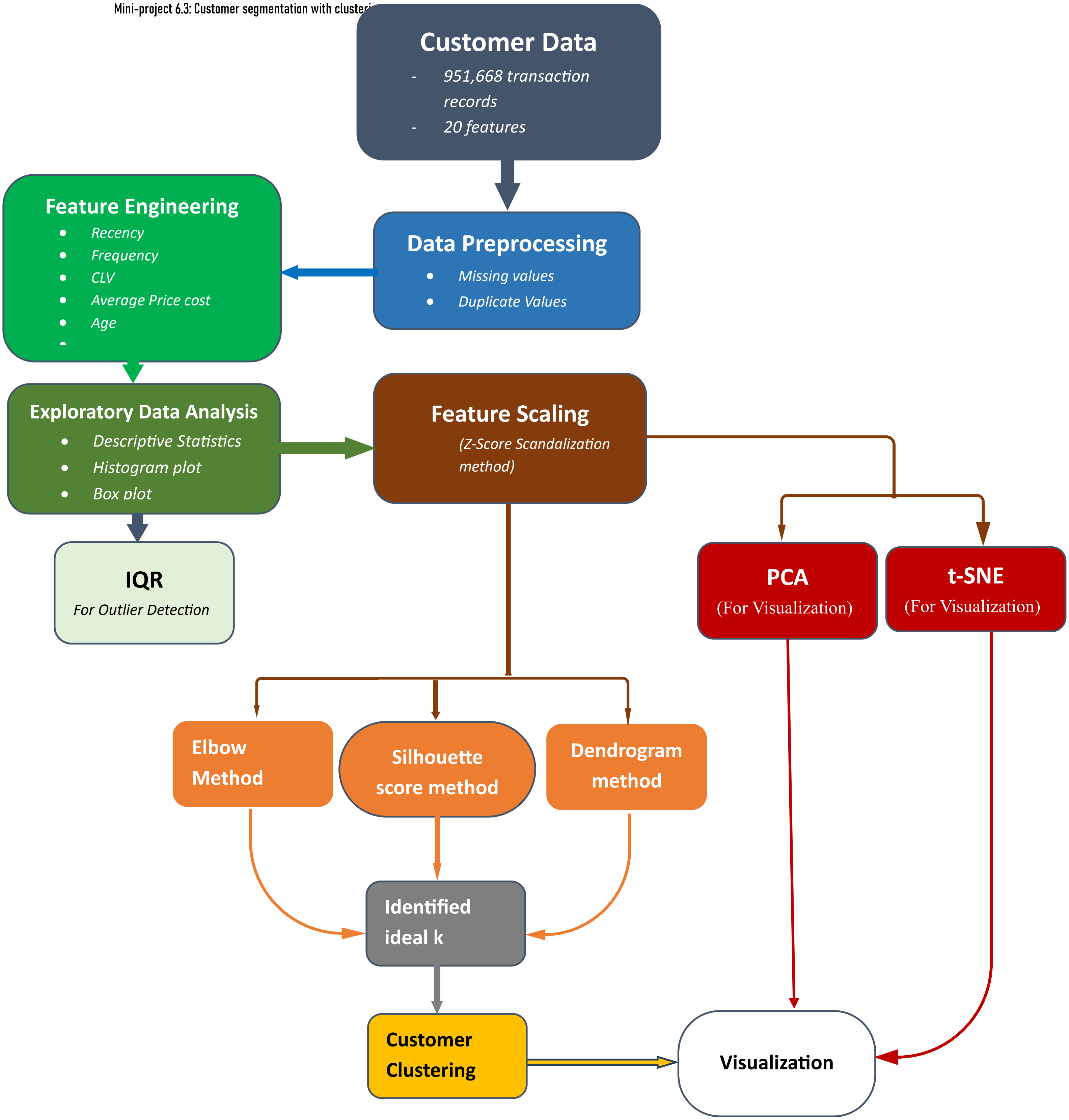

Figure 1: *Schematic Overview of Data Analysis Pipeline.*

Table 1: *Summary of data processing pipeline.*

Phase	Activity	Business Value
Preprocessing	Data cleaning and harmonization	Ensure decision are based on a clean and accurate data
Feature Engineering	Distilling 20 features into 5 Pillars	Focuses on the most customer behaviors
EDA	IQR, Boxplot & Histogram	Evaluate the presence of customer records of special interest.
Feature Scaling	Z-Score Standardization	Normalise data to avoid feature dominance.
Optimal Model & Customer segmentation	Elbow, Silhouette score, and Dendrogram	Determine the optimal k and perform k-mean Clustering
Clustering & Validation	PCA, t-SNE (for visualization)	The 5 features are condensed to 2D space to ensure cluster are distinct and reachable.

Results.

a. **Preprocessing and EDA:** Using the raw transaction data, we engineered five high-impact features for the clustering model. Exploring the features, outliers were detected using IQR method as represented on table 2.

Table 2: *Statistical Outlier Distribution Across Features*

Feature	Outlier Count	Possible Implication
Recency	3,353	Possible inactive customers
Frequency	2,654	Regular customers who are the brand ambassadors.
CLV	2,590	Possible VIPs customers representing most profitable segment.
Avg_unit_cost	2,889	Probably luxury customers that specializes in buying high-ticket items
Age	0	Suggesting that the data is clean

b. **Optimal Model:** To determine the optimal k, we adopted three methods as described on the table 3 below.

Table 3: *Different methods of determining optimal cluster number, k.*

Method	Range of clusters	Findings
Elbow method	1 & 11	k ranges between 5 & 6
Silhouette score method	2 & 11	Optimal k =5
Dendrogram method	2 -5	Optimal k = 5

i) **Elbow method:** Based on the Elbow plot generated, the optimal 'k' value for customer segmentation was determined by visually identifying the "elbow point," where the rate of decrease in the Within-Cluster Sum of Squares (WCSS) significantly slows down. From the analysis an optimal k is suggested to fall between 5 & 6 within the range of 1 to 10 clusters.

ii) **Silhouette Score method:** This method generated a range of scores corresponding with different cluster numbers. Based on the silhouette score analysis, the optimal number of clusters for customer segmentation is k = 5. This value of k is considered optimal because it yielded the highest silhouette score of approximately 0.2671 among the tested range of 2 to 10 clusters (figure 2). A higher silhouette score indicates that the clusters are well-separated from other clusters and the data points within a cluster are dense.

iii) **Dendrogram:** We sampled the scaled dataset because the notebook was crashing when trying to use the entire dataset. The horizontal dashed red line cuts at a Euclidean distance of approximately 130, effectively partitioning the data into five (5) distinct clusters (k=5). This supports the choice of k=5 for k-means clustering.

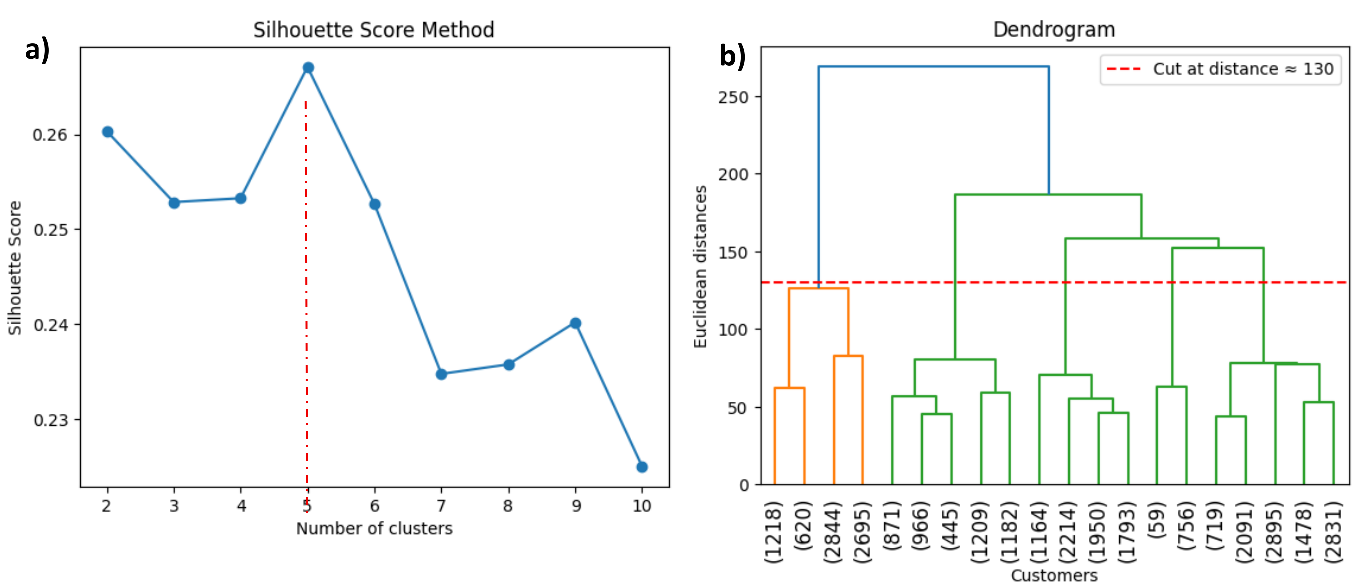

Figure 2: *Graphical representation of ideal k using (a) Silhouette method and (b) Dendrogram*

c. **Customer Segmentation:** Silhouette score and Dendrogram methods clearly established optimal value of k (number of clusters) to be equal to 5. This was fitted into the k-means model for predicting a cluster association for each customer. The customers are grouped into five (5) segments (between 0 and 4) as represented on table 4 and visualized with the aid of PCA and T-SNE (figure 3). While PCA shows broad trends, the t-SNE plot is superior at revealing local similarities and the "density" of segments (figure 5).

Table 4: *Summary of the Customer Segmentation matrix.*

Segment	Recency	CLV	Avg Unit Cost	Age	Strategic Classification
Cluster 0	**Low:** Recently active.	**Moderate:** Consistent revenue.	**Low:** Budget-conscious.	**Youngest:** Median ~37.	**Active Youth**
Cluster 1	**Low:** Highly engaged.	**Moderate-Low:** Average total spend.	**Low:** Standard pricing.	**Oldest:** Median ~67.	**Engaged Seniors**
Cluster 2	**Lowest:** Most active group.	**Highest:** Peak value "Whales".	**Low-Mid:** Volume buyers.	**Middle-Aged:** Median ~47.	**High-Value Champions**
Cluster 3	**Highest:** Longest lapse.	**Lowest:** Minimal revenue.	**Lowest:** Cheapest items.	**Mature:** Median ~57.	**Lapsed / At-Risk**
Cluster 4	**Moderate:** Occasional activity.	**High Potential:** High outliers.	**Highest:** Luxury	**Mid-Life:** Median ~52.	**Premium Shoppers**

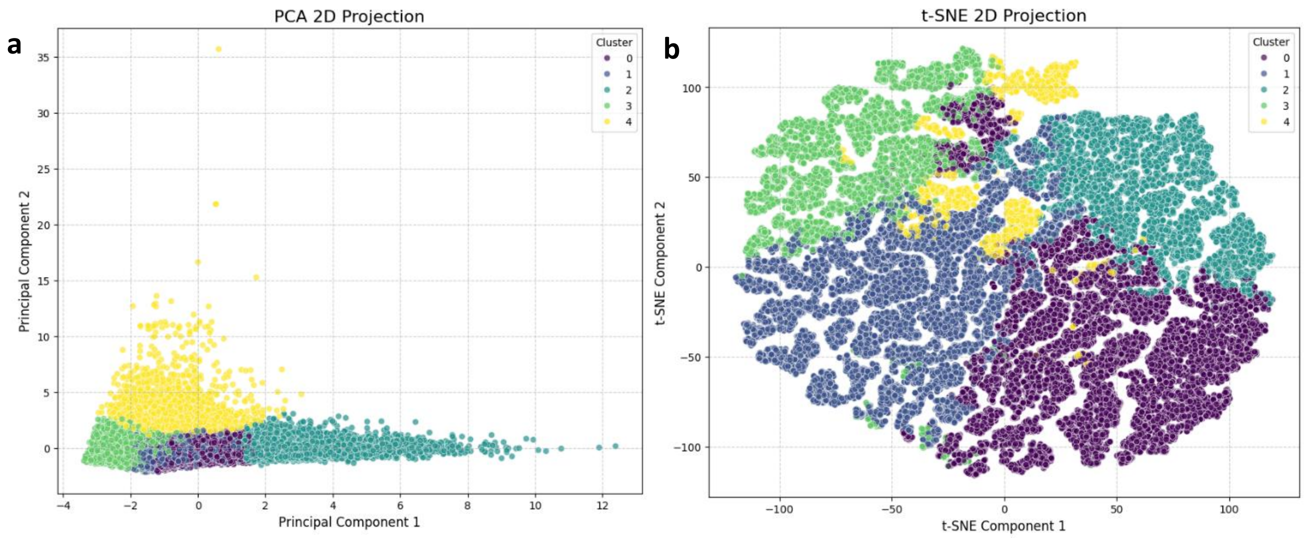

Figure 3: *2D visualization of clustering model with (a) PCA and (b) T-SNE*

Table 5: *Comparative Analysis of Dimensionality Reduction Techniques for Customer Clustering*

Feature	t-SNE View	PCA View
Clustering Quality	High quality: clusters are separately distinct but overlap at the boundaries	Moderate clustering quality: shows that segments actually overlap at their bases.
Outlier Visibility	Outliers are hidden, and are pulled into the nearest group.	Highly visible and shows exactly which clusters contain the extreme spenders.
Business Insight	Shows 5 unique "Customer Identities."	Shows Clusters that drive the most variance (profit/activity).

Conclusion and recommendation.

This report details the development of a robust customer segmentation model using a dataset of 951,668 transactions spanning five continents and 47 countries. By applying unsupervised machine learning (clustering) to core behavioral metrics, we have identified distinct customer profiles. These segments allow for a shift from "one-size-fits-all" marketing to a customer-centric strategy, optimizing budget allocation and increasing Customer Lifetime Value (CLV).

By clustering customers based on behavioral patterns rather than demographic labels, the model ensures marketing strategies remain ethical and free from demographic bias. To maintain accuracy as consumer trends and economic conditions evolve, the analysis should be updated quarterly to reflect dynamic shifts in behavior.

Retention strategy should focus on **Cluster 2** to protect the highest revenue stream through VIP membership rewards. **Cluster 4** should be targeted with exclusive, high-cost product launches, avoiding broad discount offer while win back strategy should used **for Cluster 3** customers.

Reference

SAS (2021). *CUSTOMERS_CLEAN* [Data set]. Available at: [URL/Source] (Accessed: 20 February 2024)]